

Protocol Compatibility & Normalization Matrix

Multi-Agent Ecosystem Interoperability

1. Classification & Status Vocabulary

- * LIVE: Fully active, end-to-end execution path verified against real settlement rails.
- * ADAPTER READY: Protocol normalization parser implemented and verified against ACG Canonical IR.
- * ARCHITECTURE READY / ADVISORY: Interface contract modeled for upstream/downstream advisory signals.
- * DESIGN: Architectural specification complete; awaiting industry ratification.

2. Comprehensive Compatibility Matrix

Protocol	Status	Version	Input Format	Canonical IR Mapping	Auth Mechanism	Limitation / Scope
Native ACG	LIVE	v1.0.0	Canonical Intent JSON	Direct 1:1 Schema Mapping	Ed25519 Mandate Signature	Reference protocol
REST Ingress	LIVE	v1.0.0	Standard HTTP JSON	Direct Canonical Mapping	Scoped Bearer Tokens	HTTP POST ingress
Razorpay Sandbox	LIVE	v1	Razorpay Orders API	Downstream Order Payload	Basic Auth API Keys	Sandbox environment
Model Context Protocol (MCP)	ADAPTER READY	2024-11-05	JSON-RPC tools/call	Extracts acg_checkout args	Ed25519 in arguments	HTTP transport; stdio via bridge
Agent2Agent Protocol (A2A)	ADAPTER READY	2026.1-LF	A2A Task Message	Unwraps task payload	DID + Ed25519 Mandate	Linux Foundation spec
Agentic Commerce Protocol (ACP)	ADAPTER READY	acp/1.0	Cart & Order Envelope	Maps line items & principal	Delegated Mandate	Open envelope standard
Agent Payments Protocol (AP2)	ADAPTER READY	v0.2.0	AP2 Container	Maps AP2 cart to ACG IR	Authorization Container	Authorization envelope
Universal Commerce Protocol (UCP)	ADAPTER READY	ucp-v1.2	Assistant Journey Request	Maps order_lines to catalog	Delegated Mandate	Journey checkout format
Visa Trusted Agent Protocol (TAP)	DESIGN	tap/1.0-draft	Hardware Attestation Container	Maps TEE token & items	Enclave Hardware Attestation	Specification stage
Razorpay Vulcan Intelligence	ARCHITECTURE READY	vulcan-v1.4	Transaction Context	Advisory routing hints	Non-authoritative Advisory	Downstream intelligence only